

ECLAC-MTTCN: Environment-Conditioned Attention over Multi-Zone Temporal Tokens for 10-Minute-Ahead Load Forecasting

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Abstract—This study evaluates ECLAC-MTTCN, a joint three-zone neural forecaster for the next 10-minute electricity-demand observation. The model receives the preceding 144 observations (24 hours) of three zone loads and five environmental variables. Three independent temporal convolutional encoders represent zone histories, a fourth encoder represents environmental history, and four-head additive attention scores all 432 zone-time tokens using the latest target-zone and environmental states. Feature-wise fusion combines local, attended multi-zone, and environmental representations before zone-specific residual heads predict corrections to persistence. On a chronological 7,863-sample test partition of the Tetouan City dataset, the 365,315-parameter model achieves macro MAE 191.7754, RMSE 296.6900, R^2 0.9961, and MAPE 0.9909%. It reduces MAE and RMSE by 34.57% and 34.03%, respectively, relative to persistence and outperforms the implemented plain TCN, LSTM, and GRU baselines. A fixed- α ridge regressor remains better on all four macro metrics (MAE 187.1133 and RMSE 287.1467), so the results do not support overall superiority. Four ablations identify residual correction as the most consequential component; notably, equal-weight fusion slightly outperforms the adaptive gate in this single run. Attention, fusion, and paired-test results are therefore interpreted descriptively, with conclusions limited to one dataset split and one seed.

Index Terms—short-term load forecasting, temporal convolutional network, multi-zone forecasting, attention, adaptive fusion, Tetouan City

I. INTRODUCTION

Accurate short-term load forecasting supports scheduling, load balancing, and operational planning in distribution systems. Urban distribution zones are neither independent nor identical: they share weather and activity patterns while retaining zone-specific levels and temporal dynamics. The Tetouan City Power Consumption dataset provides 10-minute observations from three distribution zones together with temperature, humidity, wind speed, general diffuse flows, and diffuse flows [1]. It has been used to compare conventional machine-learning methods [2], deep residual and recurrent architectures [3], shift-based forecasting formulations [4], and recent multi-model benchmarking frameworks [5].

The implementation studied here combines four ideas: causal temporal convolution, environment-conditioned attention over multi-zone temporal states, feature-wise three-source fusion, and residual forecasting around persistence. The intended motivation is to retain target-zone dynamics while

allowing the prediction head to use temporally indexed states from all zones and current environmental context. However, architectural complexity must be judged against strong simple baselines, especially at a 10-minute horizon where persistence and linear lag models can be competitive.

The contributions supported by the executable notebook and its stored outputs are as follows:

- A leakage-conscious chronological pipeline uses training-only standardization and 144-step histories containing only observations available at the forecast origin.
- The implemented ECLAC-MTTCN architecture is specified mathematically, including its 29-step TCN receptive field, attention over 432 temporal tokens, feature-wise fusion, and persistence-centered decoder.
- Persistence, fixed- α ridge, plain TCN, LSTM, and GRU baselines are compared on the same recorded test targets, together with four implemented ablations.
- Descriptive attention and fusion diagnostics are reported without treating internal weights as causal explanations.

The central empirical result is deliberately qualified: ECLAC-MTTCN improves on persistence and the tested deep baselines, but ridge gives the strongest aggregate accuracy.

II. RELATED WORK

Temporal convolutional networks (TCNs) use causal dilated convolutions and residual blocks to model sequential data while permitting parallel training [6]. TCN variants have been applied to short-term load forecasting, including architectures that combine convolutional transformations with input attention [7]. Attention mechanisms form data-dependent weighted combinations of sequence states [8]; in multi-zone forecasting, their weights can summarize allocation across source zones and historical positions, but they do not establish causal influence.

Prior Tetouan studies show that both nonlinear and conventional models can be competitive. Salam and El Hibaoui compared several machine-learning algorithms on the dataset [2]; later work investigated deep residual and recurrent models [3] and shifted-consumption formulations [4]. A recent Darts-based comparative study evaluated multiple statistical, machine-learning, and deep models at 10-minute and hourly resolutions [5]. These findings motivate a high-dimensional

TABLE I
CHRONOLOGICAL DATA PARTITION

Partition	Rows	Samples	Timestamp range
Train	36,691	36,547	Jan. 1–Sep. 12, 19:00
Validation	7,862	7,862	Sep. 12, 19:10–Nov. 6, 09:20
Test	7,863	7,863	Nov. 6, 09:30–Dec. 30, 23:50

ridge comparator rather than evaluation only against persistence and recurrent networks. The present work is narrower than multi-horizon or probabilistic studies: it evaluates deterministic one-step-ahead forecasts and does not estimate predictive uncertainty.

III. METHODOLOGY

A. Forecasting Task and Dataset

Let $\mathbf{x}_t \in \mathbb{R}^3$ denote the three zone loads observed at forecast origin t , and let $\mathbf{e}_t \in \mathbb{R}^5$ denote the environmental vector. With lookback $L = 144$, the model input and target are

$$\begin{aligned}\mathbf{X}_t^x &= [\mathbf{x}_{t-L+1}, \dots, \mathbf{x}_t] \in \mathbb{R}^{L \times 3}, \\ \mathbf{X}_t^e &= [\mathbf{e}_{t-L+1}, \dots, \mathbf{e}_t] \in \mathbb{R}^{L \times 5}, \\ \hat{\mathbf{x}}_{t+1} &= \mathcal{F}_\theta(\mathbf{X}_t^x, \mathbf{X}_t^e, \mathbf{x}_t).\end{aligned}\quad (1)$$

Thus, 24 hours of observed history are used to forecast all three loads 10 minutes ahead. DateTime is used for sorting, validation, partitioning, and plotting, but is excluded from the model inputs.

The executed code loads 52,416 complete records from 1 January to 30 December 2017. It rejects duplicate timestamps, missing or non-finite model values, and intervals other than 10 minutes; no imputation is performed. The official UCI record describes the three distribution-network targets and the five environmental variables used here [1]. Fig. 1 shows the hourly-averaged load profiles for visualization only; training retains the native 10-minute resolution.

B. Chronological Split, Scaling, and Windows

Rows are assigned chronologically to 70% training, 15% validation, and 15% test target ranges. Separate StandardScaler instances for the three load variables and five environmental variables are fitted only on training rows. For variable j ,

$$\tilde{x}_{t,j} = \frac{x_{t,j} - \mu_j^{\text{tr}}}{\sigma_j^{\text{tr}}}, \quad (2)$$

where both parameters are calculated from the training rows. The fitted transformations are then applied unchanged to validation and test rows.

Windows are formed from the full ordered arrays after target ranges are assigned. Therefore, the first validation target may use the final 144 training observations as history, and the first test target may use the final 144 validation observations. Every input nevertheless precedes its target, and no future row enters a window. Table I reports the exact executed split.

Training-only diagnostics support the multi-zone formulation without serving as feature selection. Contemporaneous

Pearson correlations among zone loads are 0.92 (Zones 1–2), 0.82 (Zones 1–3), and 0.85 (Zones 2–3). Lead-lag curves also recover strong daily recurrence near 24 hours. These are descriptive correlations and are not interpreted causally.

C. Causal Temporal Encoders

The architecture contains three independent single-channel zone encoders and one five-channel environmental encoder. Each maps its input to hidden width $D = 64$ using a 1×1 convolution, followed by three residual blocks with dilations $d \in \{1, 2, 4\}$. Each block contains two causal one-dimensional convolutions of kernel size $k = 3$, channel-wise layer normalization at each time index, GELU activation, and dropout 0.15. If $\mathbf{H}^{(b-1)}$ is the input to block b , then

$$\mathbf{H}^{(b)} = \text{operatorname{GELU}}\left(\mathbf{H}^{(b-1)} + \mathcal{B}_{d_b}(\mathbf{H}^{(b-1)})\right), \quad (3)$$

where $\mathcal{B}_{d_b}$ denotes the normalized, activated, and regularized two-convolution branch. A final layer normalization follows the stack. Its receptive field is

$$R = 1 + 2(k-1) \sum_{d \in \{1, 2, 4\}} d = 29 \quad (4)$$

samples, equivalent to 4 hours 50 minutes. This is the receptive field of each encoded state; attention still receives all 144 encoded time indices.

D. Environment-Conditioned Multi-Zone Attention

Let $\mathbf{h}_{z,\ell} \in \mathbb{R}^{64}$ be the zone-encoder state for zone $z \in \{1, 2, 3\}$ and temporal index $\ell \in \{1, \dots, L\}$, and let $\mathbf{p}_t \in \mathbb{R}^{64}$ be the final environmental state. The latest target-zone state forms the query. In head r , whose dimension is $d_h = 16$,

$$\begin{aligned}\mathbf{q}_z^r &= W_q^r \mathbf{h}_{z,L}, & \mathbf{k}_{m,\ell}^r &= W_k^r \mathbf{h}_{m,\ell}, \\ \mathbf{u}^r &= W_e^r \mathbf{p}_t, & s_{z,m,\ell}^r &= \frac{(\mathbf{a}^r)^\top \tanh(\mathbf{q}_z^r + \mathbf{k}_{m,\ell}^r + \mathbf{u}^r)}{\sqrt{d_h}}.\end{aligned}\quad (5)$$

The scores are normalized across all $3L = 432$ zone-time tokens:

$$\begin{aligned}\alpha_{z,m,\ell}^r &= \frac{\exp(s_{z,m,\ell}^r)}{\sum_{m'=1}^3 \sum_{j=1}^L \exp(s_{z,m',j}^r)}, \\ \mathbf{o}_z^r &= \sum_{m=1}^3 \sum_{\ell=1}^L \alpha_{z,m,\ell}^r W_v^r \mathbf{h}_{m,\ell}.\end{aligned}\quad (6)$$

The four contexts are concatenated and projected; dropout, a local residual connection, and layer normalization produce the attended representation. During training, dropout is applied to the normalized weights and projected context; it is disabled during evaluation.

The implementation includes the target zone's own 144 tokens and has no self-zone mask. It also contains no positional or relative-lag embedding. Consequently, the module attends to temporally indexed states, but it is not an explicitly lag-encoded or exclusively cross-zone attention mechanism. This distinction is retained throughout the analysis.

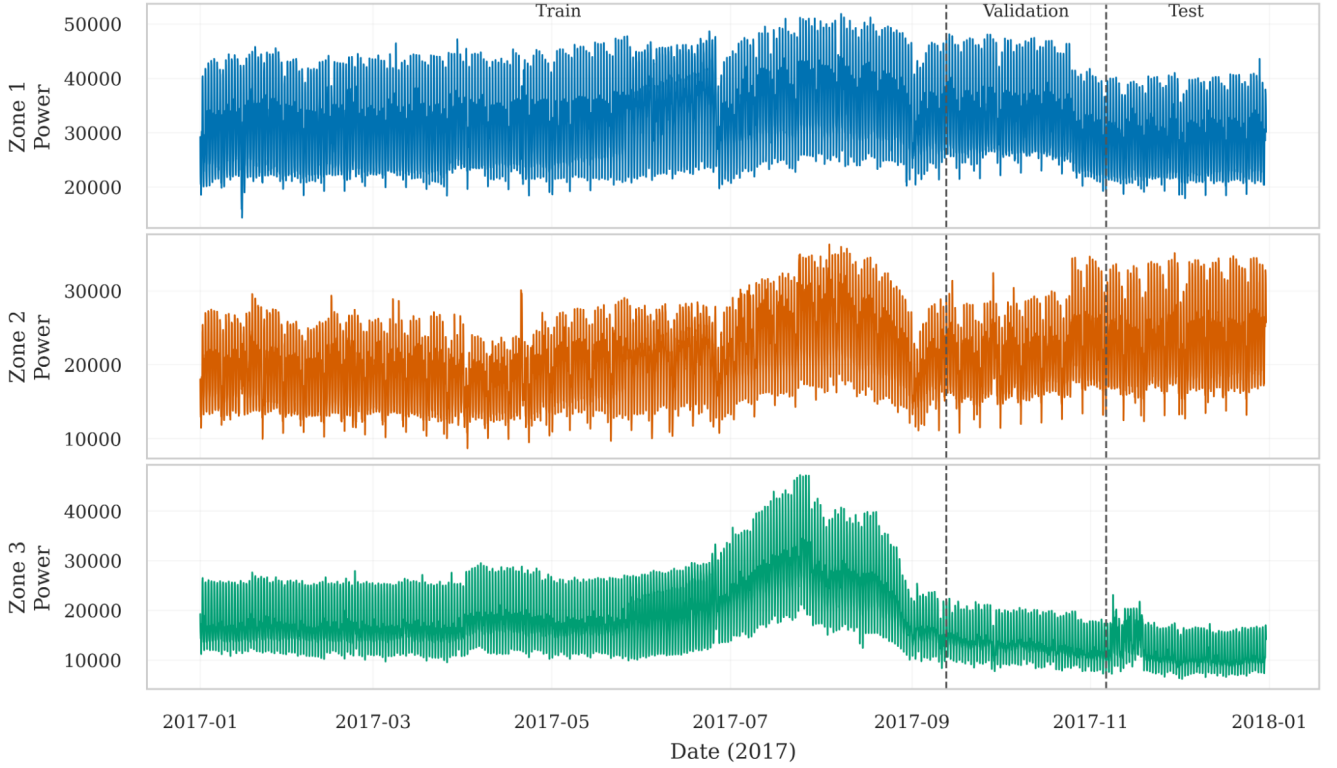


Fig. 1. Hourly-averaged load profiles with the chronological train, validation, and test boundaries. Aggregation is used only in this visualization; all model inputs and targets remain at 10-minute resolution.

E. Feature-Wise Fusion and Residual Decoder

For target zone z , linear projections produce local $\mathbf{l}_z$, attended $\mathbf{a}'_z$, and environmental $\mathbf{e}'_z$ vectors in $\mathbb{R}^{64}$. A two-layer gate network maps their concatenation to three logits for every hidden feature. For feature j ,

$$[g_{z,j}^{(l)}, g_{z,j}^{(a)}, g_{z,j}^{(e)}] = \text{softmax}_3(\mathcal{G}([\mathbf{l}_z; \mathbf{a}'_z; \mathbf{e}'_z])_j), \quad (7)$$

$$\mathbf{f}_{z,j} = g_{z,j}^{(l)} \mathbf{l}_{z,j} + g_{z,j}^{(a)} \mathbf{a}'_{z,j} + g_{z,j}^{(e)} \mathbf{e}'_{z,j}. \quad (8)$$

A zone-specific $64 \rightarrow 32 \rightarrow 1$ head predicts a standardized residual correction:

$$\hat{\tilde{x}}_{z,t+1} = \tilde{x}_{z,t} + \rho_z(\mathbf{f}_z). \quad (9)$$

The last linear layer of every residual head is initialized to zero, so the initial network reproduces persistence. Predictions are inverse-transformed before evaluation. Fig. 2 summarizes the implemented data flow.

F. Optimization and Reproducibility

Training minimizes PyTorch's mean Huber loss with $\delta = 1$ [9]. AdamW uses learning rate 10^{-3} and weight decay 10^{-4} [10]; the global gradient norm is clipped to 1.0. Batch size is 64, the maximum is 40 epochs, and early-stopping patience is 10 validation epochs. ReduceLROnPlateau uses factor 0.5, configured patience 3, and minimum learning rate 10^{-6} . Automatic mixed precision is enabled on CUDA, and the lowest-validation-loss checkpoint is restored.

The recorded full run uses seed 42 for Python, NumPy, and PyTorch; CUDA generators are seeded, cuDNN benchmarking is disabled, and deterministic mode is enabled. It ran with Python 3.12.13, PyTorch 2.11.0+cu128, scikit-learn 1.6.1, and an NVIDIA Tesla T4. The proposed model stopped after epoch 28 and restored epoch 18, whose validation Huber loss was 0.000820 (Fig. 3). It has 365,315 trainable parameters: 300,800 in the four encoders, 20,736 in attention, 37,440 in fusion, and 6,339 in the three heads.

G. Baselines and Ablations

Persistence copies the latest observed load. Ridge regression uses one multi-output estimator with fixed $\alpha = 1$ and all $144(3 + 5) = 1,152$ flattened predictors; α is not validation-tuned [11]. The plain TCN concatenates all eight input channels and uses the same hidden width and dilation schedule as one ECLAC-MTTCN encoder. The LSTM [12] and GRU [13] are one-layer, width-64 models. Each deep baseline predicts a persistence residual and follows the same optimizer-level training routine.

The four implemented ablations are listed in Table II. In the no-environment variant, the environmental encoder still executes, but its final state is replaced by zeros before attention and fusion; downstream projection biases remain. The no-temporal-token-attention variant replaces the custom module with four-head multihead attention over only the latest three zone states. It therefore changes token coverage, scoring type, and environmental conditioning together and does not

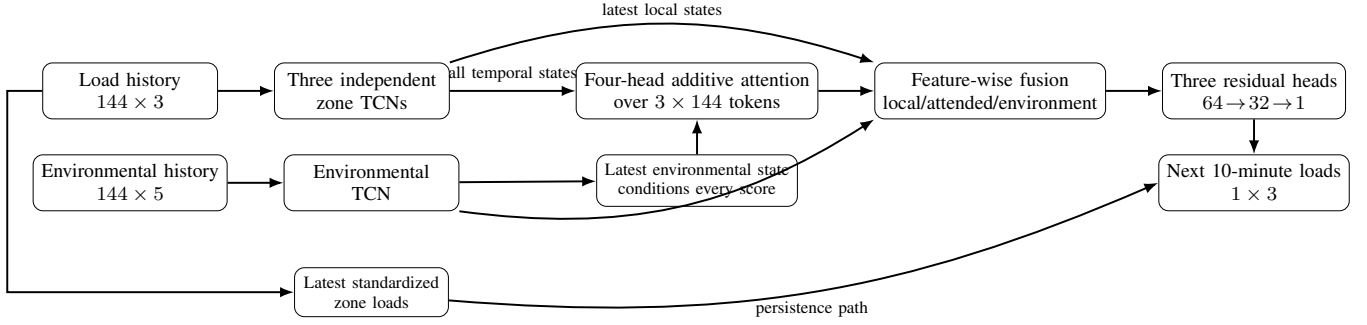


Fig. 2. Implementation-aligned ECLAC-MTTCN architecture. Attention scores all 432 zone-time tokens, including same-zone tokens, and is conditioned on the latest environmental state. The decoder adds learned residuals to the latest standardized loads.

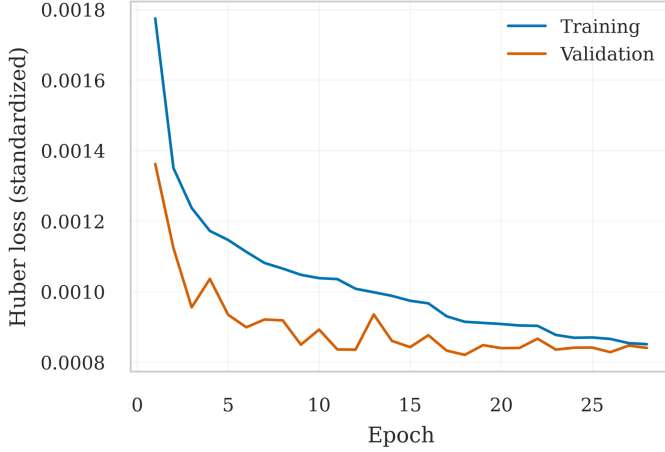


Fig. 3. Recorded training and validation Huber losses. Early stopping occurred after epoch 28, and the minimum-validation checkpoint from epoch 18 was restored.

TABLE II
IMPLEMENTED ABLATIONS

Variant	Implemented change
No environment	Zero environmental summary before downstream use.
No temporal-token attention	Attend only over the latest three zone states using standard MHA.
No adaptive gate	Use equal 1/3 weights for the three projected sources.
No residual	Predict standardized loads directly; do not zero-initialize the final layer.

isolate a pure lag effect. No separate no-cross-zone ablation is implemented.

H. Evaluation and Exploratory Test

All predictions are inverse-transformed before metric calculation. For one zone with N test targets,

$$\begin{aligned}
 \text{MAE} &= \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|, \\
 \text{RMSE} &= \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}, \\
 R^2 &= 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}, \\
 \text{MAPE} &= \frac{100}{N} \sum_{i=1}^N \frac{|y_i - \hat{y}_i|}{\max(|y_i|, 10^{-8})}.
 \end{aligned} \tag{10}$$

Each metric is first computed independently for each zone; the reported macro value is the unweighted mean of the three zone metrics.

The notebook additionally applies a one-sided Wilcoxon signed-rank test [14] to paired absolute errors for each zone. With $d_i = |e_i^{\text{ECLAC-MTTCN}}| - |e_i^{\text{pers}}|$, the alternative is that the distribution of d_i is shifted below zero. The implementation uses `zero_method=wilcox`. No serial-dependence or multiplicity correction is applied, so the test is reported as exploratory.

IV. RESULTS AND DISCUSSION

A. Predictive Accuracy

Table III reports macro test performance. ECLAC-MTTCN reduces persistence MAE by 34.57% and RMSE by 34.03%, and it outperforms the implemented plain TCN, LSTM, and GRU on all four macro metrics. Among the deep baselines, GRU is strongest, but its MAE and RMSE remain 2.51% and 1.76% above those of ECLAC-MTTCN, respectively.

Ridge is the strongest aggregate model. Relative to ridge, ECLAC-MTTCN has 2.49% higher MAE, 3.32% higher RMSE, 2.36% higher MAPE, and a slightly lower R^2 . This outcome demonstrates why the linear baseline cannot be omitted and prevents a state-of-the-art or overall-superiority claim. Fig. 4 visualizes the zone-level RMSE comparison.

TABLE III
MACRO-AVERAGED TEST PERFORMANCE ACROSS THREE ZONES

Model	MAE ↓	RMSE ↓	R^2 ↑	MAPE (%) ↓
Persistence	293.1133	449.7262	0.9915	1.4609
Ridge ($\alpha = 1$)	187.1133	287.1467	0.9963	0.9681
ECLAC-MTTCN	191.7754	296.6900	0.9961	0.9909
Plain TCN	212.3534	314.4221	0.9954	1.1507
LSTM	222.3424	328.8288	0.9950	1.1859
GRU	196.5956	301.9046	0.9958	1.0338

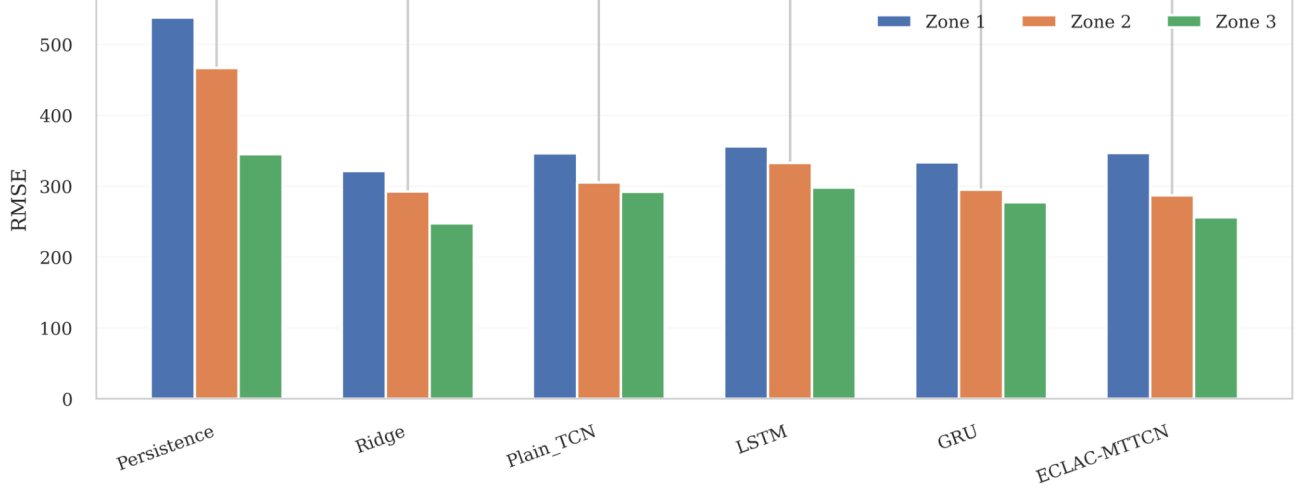


Fig. 4. Zone-level RMSE for the executed persistence, ridge, deep-learning, and ECLAC-MTTCN runs. Ridge has the lowest RMSE in Zones 1 and 3, whereas ECLAC-MTTCN has the lowest Zone 2 RMSE.

Table IV shows that the ranking is not uniform across zones. Ridge is better on all four metrics in Zone 1. In Zone 2, ridge has lower MAE and MAPE, while ECLAC-MTTCN has lower RMSE (287.1055 versus 292.6393) and marginally higher R^2 . In Zone 3, ECLAC-MTTCN has lower MAE (144.3554 versus 146.8795), while ridge is better on RMSE, R^2 , and MAPE.

Fig. 5 plots the first 1,008 chronological test targets, corresponding to one week. The interval is fixed before inspection rather than selected post hoc. The proposed trajectories closely follow the actual series, although sharp Zone 3 events remain visibly more difficult. All reported metrics use the complete 7,863-target test partition.

The absolute-error empirical cumulative distributions in Fig. 6 confirm that ridge and ECLAC-MTTCN substantially dominate persistence over most of each zone’s error distribution. Ridge is generally shifted slightly left of ECLAC-MTTCN in Zones 1 and 2; the two are closely matched in Zone 3. The long right tails, particularly in Zone 2, show why RMSE and MAE should be reported together.

B. Ablation Analysis

Table V reports the four executed ablations. Direct prediction without persistence residuals produces the largest degradation: MAE increases by 60.54%, RMSE by 39.89%, and MAPE by 81.29%. Removing the environmental summary has a small adverse effect (MAE +0.99%, RMSE +0.05%), while replacing full temporal-token attention with latest-state MHA

increases MAE by 1.54% and RMSE by 1.43%. Because that ablation changes several attention properties simultaneously and every variant is trained only once, these differences are suggestive rather than component-wise causal estimates.

The adaptive gate is not supported by this run’s ablation. Equal 1/3 fusion improves MAE by 1.57%, RMSE by 0.95%, and MAPE by 0.41% relative to the full model. Consequently, the present evidence supports the residual decoder and provides limited support for environmental and full-history attention, but it does not demonstrate a forecasting advantage from learned adaptive fusion. Repeated-seed experiments would be required to determine whether the small differences are stable.

C. Descriptive Attention and Fusion Behavior

Attention is averaged over test samples and heads, then summed over the 144 positions to obtain source-zone mass. As shown in Fig. 7, target Zone 1 allocates masses 0.403, 0.254, and 0.343 to source Zones 1–3; the corresponding rows for target Zones 2 and 3 are (0.392, 0.261, 0.346) and (0.390, 0.248, 0.362). Same-zone attention is therefore substantial, consistent with the absence of a self-zone mask. Detailed temporal heatmaps place the largest mean weights near the oldest portion of the 24-hour window, approximately 22 hours before the origin. Since no explicit lag encoding is implemented, this pattern is an observed allocation of the learned encoded states, not proof of a causal or periodic mechanism.

TABLE IV
ZONE-LEVEL TEST PERFORMANCE FOR PRINCIPAL COMPARATORS

Zone	Model	MAE	RMSE	R^2	MAPE (%)
1	Persistence	364.3652	537.8128	0.9921	1.2703
	Ridge	228.6876	321.2141	0.9972	0.8060
	ECLAC-MTTCN	236.9714	346.7885	0.9967	0.8379
2	Persistence	312.3973	466.4549	0.9930	1.3555
	Ridge	185.7728	292.6393	0.9973	0.8098
	ECLAC-MTTCN	193.9993	287.1055	0.9974	0.8419
3	Persistence	202.5774	344.9107	0.9895	1.7568
	Ridge	146.8795	247.5868	0.9946	1.2887
	ECLAC-MTTCN	144.3554	256.1759	0.9942	1.2928

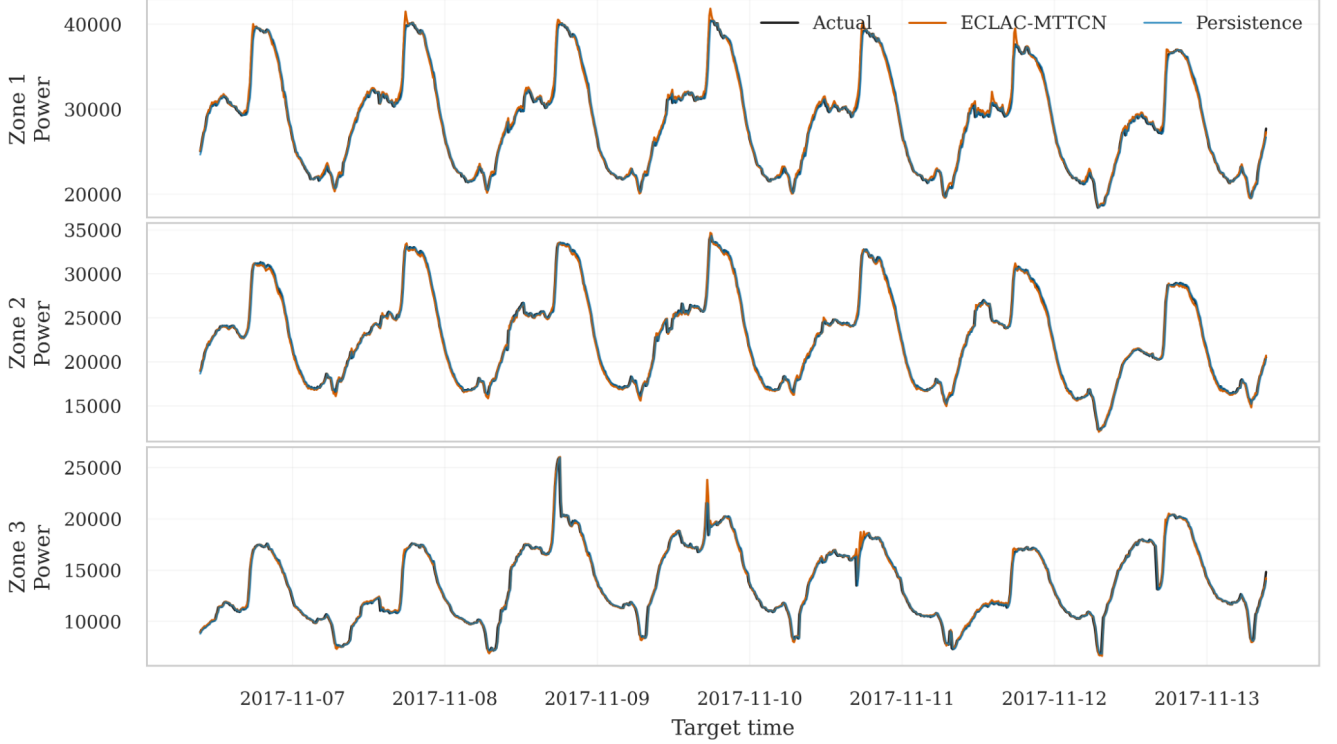


Fig. 5. Actual load, ECLAC-MTTCN, and persistence for the first 1,008 chronological test targets (one week). The plot is illustrative; quantitative results use all 7,863 test samples.

The feature-level gates are averaged across 64 features and then across test samples. Mean local/attended/environmental allocations are $(0.3398, 0.3917, 0.2685)$ for Zone 1, $(0.3496, 0.3761, 0.2743)$ for Zone 2, and $(0.3444, 0.3656, 0.2900)$ for Zone 3. The attended component receives the largest mean allocation in every zone. Nevertheless, the no-gate ablation performs slightly better, illustrating that a larger internal gate value does not establish predictive necessity.

D. Exploratory Paired Tests

The stored Wilcoxon tests compare ECLAC-MTTCN only with persistence. All three one-sided p -values are extremely small, alongside zone-level MAE reductions from 28.74% to 37.90% (Table VI). However, adjacent 10-minute errors are

serially dependent, and the implementation applies neither a dependence-aware correction nor a multiplicity adjustment. The values document strong paired evidence against persistence within this particular test sequence, but they do not replace evaluation across independent forecast origins, datasets, or seeds. No paired test against ridge is implemented.

V. LIMITATIONS AND REPRODUCIBILITY

The experiment covers one city, one calendar year, one chronological holdout, one random seed, and one 10-minute-ahead horizon. It does not implement rolling-origin evaluation, hyperparameter search, repeated-seed uncertainty, probabilistic forecasts, calibration, MASE, sMAPE, or peak-specific metrics. Ridge is fixed rather than tuned, yet it remains the best

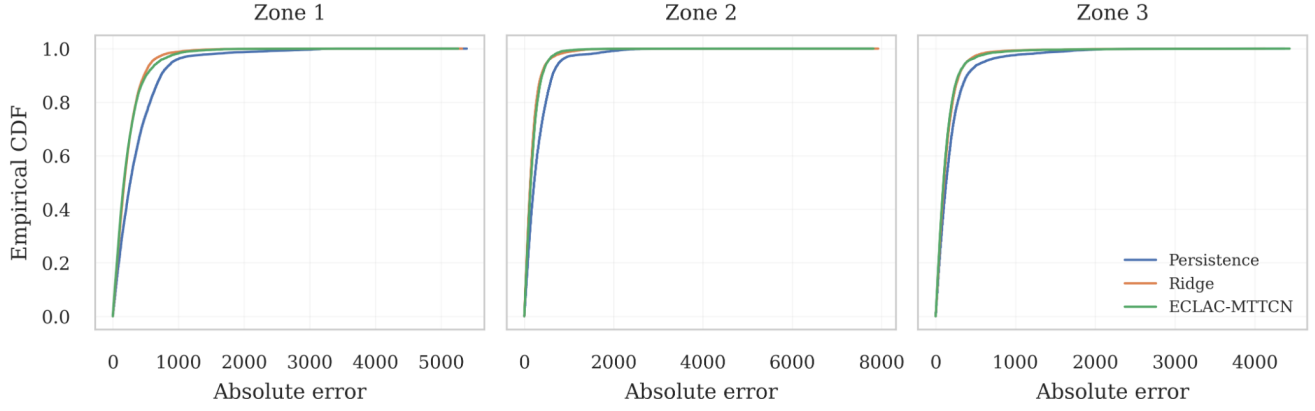


Fig. 6. Empirical CDFs of absolute test error for persistence, ridge, and ECLAC-MTTCN. Curves use the complete test partition. A curve farther left/up indicates smaller errors.

TABLE V
MACRO-AVERAGED ABLATION RESULTS

Variant	MAE	Δ MAE (%)	RMSE	Δ RMSE (%)	R^2	MAPE (%)
Full ECLAC-MTTCN	191.7754	—	296.6900	—	0.9961	0.9909
No environment	193.6735	+0.99	296.8382	+0.05	0.9961	1.0043
No temporal-token attention	194.7198	+1.54	300.9268	+1.43	0.9958	1.0246
No adaptive gate	188.7647	−1.57	293.8859	−0.95	0.9961	0.9868
No residual	307.8847	+60.54	415.0503	+39.89	0.9910	1.7964

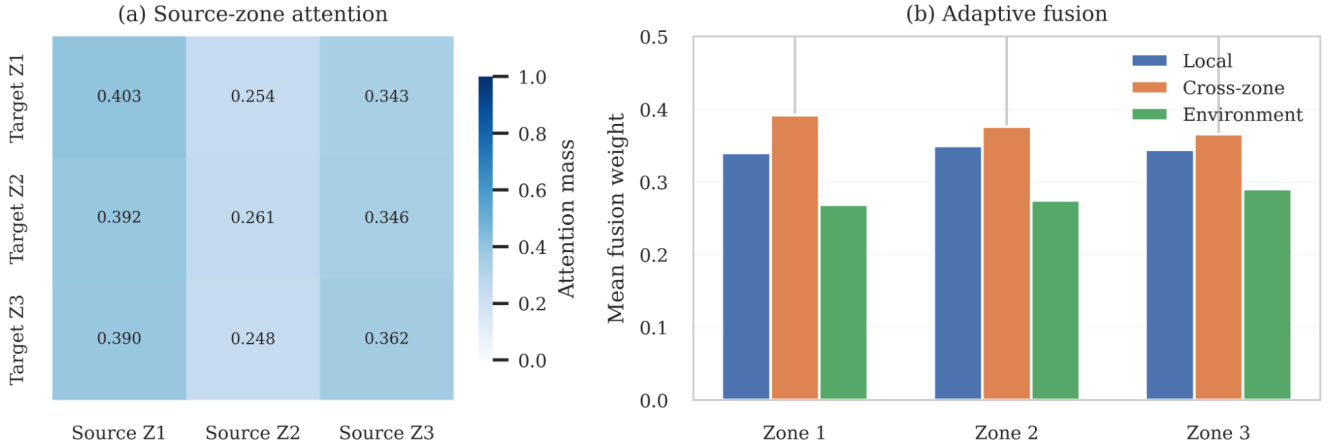


Fig. 7. Test-set interpretability summary. (a) Mean source-zone attention mass integrated over 144 temporal positions; diagonal cells are nonzero because same-zone tokens are included. (b) Mean feature-averaged fusion weights. These internal allocations are descriptive and not causal effects.

TABLE VI
EXPLORATORY WILCOXON TESTS VS. PERSISTENCE

Zone	Statistic	One-sided p	MAE red. (%)
1	9,903,064.5	5.68×10^{-168}	34.96
2	9,661,326.5	1.07×10^{-182}	37.90
3	8,987,400.5	4.69×10^{-227}	28.74

macro model. Broader or state-of-the-art superiority claims are therefore unsupported.

The attention mechanism has no positional embedding and includes same-zone tokens; attention weights cannot identify

causal cross-zone effects. The no-temporal-token ablation changes more than one mechanism, the no-environment variant retains encoder execution and downstream biases, and no isolated no-cross-zone variant exists. Adaptive fusion is not validated by the current ablation. Moreover, the shared seeded training DataLoader generator is not recreated for every extended neural experiment, so the candidates share a split and initial seed but are not guaranteed identical mini-batch permutations.

The notebook records the main software versions and GPU, but it does not measure training duration, peak memory, or inference latency. Future work should add repeated seeds, rolling

forecast origins, dependence-aware comparisons against ridge, an isolated cross-zone ablation, and computational profiling. Reproduction should preserve the exact split, training-only scalers, 144-step window, and recorded constants.

VI. CONCLUSION

This implementation-aligned study evaluates a 365,315-parameter model for joint three-zone, 10-minute-ahead load forecasting from 24 hours of load and environmental history. Independent causal TCN encoders, environment-conditioned attention over 432 temporal tokens, feature-wise fusion, and persistence residual heads yield substantial gains over persistence and the executed plain TCN, LSTM, and GRU baselines. Ridge remains better on every macro metric, while ECLAC-MTTCN wins selected zone-level comparisons. Ablations identify residual correction as the dominant supported component, show small benefits from environment and temporal-token attention under the recorded run, and provide no accuracy support for adaptive gating. The model is therefore best characterized as a competitive and diagnostically transparent experimental architecture under this split, not as a generally superior forecasting system.

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